Solutions: Test A – Foundation (topics 01–04)

Pass mark 75 %: 18 out of 24 points Only look here once you have finished the test. You grade practical tasks and safety checks yourself using the self-assessment rubric (`self_assessment`).

Topics 01 – What is AI?

Cheatsheet: `01_cheatsheets/01_what_is_ai`

Question 1 – B (1 pt): A language model predicts likely-fitting pieces of text. It does not look up verified knowledge (A), does not think like a human (C) and is not a search engine (D).

Question 2 – False (1 pt): A language model produces what is probable, not what is verified. Convincing style says nothing about correctness. Remember: "Sounding convincing does not mean it's true."

Question 3 – A, B, E (2 pts): Rephrasing, summarizing and collecting ideas are strengths. Current events (C) without web search and exact sources (D) are among the weaknesses. You have to check them.

Question 4 – C (1 pt): A chatbot generates answers anew from probabilities. A classic search engine shows existing pages. Some chatbots can additionally search, but their answer is still, first and foremost, generated text.

Question 5 – D (1 pt): Without web search, the AI may not know current things and guesses. Web search or cross-checking with a reliable source is right. Asking repeatedly (B) only yields the same uncertainty multiple times.

Topic 02 – Tokens & context window

Cheatsheet: `01_cheatsheets/02_tokens_context`

Question 6 – A (1 pt): Tokens are small pieces of text in which the AI processes text.

Question 7 – False (1 pt): The context window is the chat's working memory. It also contains the conversation history, files, instructions and the answer.

Question 8 – B (1 pt): Summarizing, checking and continuing in a new chat clears out the "context junk". Capital letters (A) and waiting (C, D) do not solve the problem.

Question 9 – Model answer (2 pts):

```
Summarize our chat so far in at most 10 lines.
Include: decisions made, my instructions and open questions.
I will copy the summary into a new chat.
```

- Criterion (a) 1 pt: length limited (e.g., "at most 10 lines").
- Criterion (b) 1 pt: important content named (e.g., decisions, instructions, open questions).

Topic 03 – Model selection & free plans

Cheatsheet: `01_cheatsheets/03_model_selection`

Question 10 – C (1 pt): For simple tasks, a smaller, faster model is enough. The largest model with reasoning effort (A) unnecessarily costs limit.

Question 11 – A, B, C (2 pts): Free plans often restrict models, usage amount and features. They can fundamentally write texts (D). Anonymizing (E) is always allowed.

Question 12 – False (1 pt): Larger models are often slower and use more limit. For simple tasks they add little. They make mistakes too.

Question 13 – A (1 pt): For consequential decisions, the rule is: larger model and check the result against the original. Adopting unchecked (B, D) or asking repeatedly (C) is no substitute for checking.

Question 14 – 1 K, 2 K, 3 G, 4 G (2 pts): Translating and correcting are simple tasks. Weighing multiple conditions and finding contradictions in long texts are tricky. - 4 correct = 2 points, 3 correct = 1 point. - Note: This is a rule of thumb. What matters is that you can justify your assignment.

Topic 04 – Data privacy in everyday life

Cheatsheet: `01_cheatsheets/04_data_privacy_everyday`

Question 15 – D (1 pt): Fictional data with placeholders is unproblematic. ID card (A), customer data (B) and passwords (C) never belong in a chat.

Question 16 – Model answer (2 pts): (a) 1 pt if you name at least three: name, address, phone number, health detail, and possibly the company name. (b) 1 pt if the task remains recognizable and placeholders are used instead of real data:

```
Write a friendly rejection letter to Person A, who applied as an
accountant at Company X. The tone should be appreciative and brief.
```

The health detail is omitted, because it is not needed for the task.

Question 17 – False (1 pt): Screenshots often contain more: background, other tabs, names, metadata. Crop or black out.

Question 18 – B (1 pt): With work data, first clarify what is allowed and use the company account if one exists. Leaving out names (C) is not enough, because confidential content remains. Deleting afterwards (D) does not undo the upload.

Question 19 – A (1 pt): Turning off training use reduces the risk, if the service offers this option. A word like “confidential” (B) has no effect. Not every provider has the same options. Check your service.

Counting your points

Topic	Questions	Possible points	My points
01 What is AI?	1–5	6	
02 Tokens & context window	6–9	5	
03 Model selection & free plans	10–14	7	
04 Data privacy in everyday life	15–19	6	
Total		**24**	

Passed: 18 points or more.

What to review if you got something wrong

- **Wrong on questions 1–5:** Topic 01 *What is AI?* (also topic 08 *Hallucinations*, once you have worked through it)
- **Wrong on questions 6–9:** Topic 02 *Tokens & context window* and extra exercises 02
- **Wrong on questions 10–14:** Topic 03 *Model selection & free plans* and extra exercises 03
- **Wrong on questions 15–19:** Topic 04 *Data privacy in everyday life* and extra exercises 04

Not passed? Review the topics with the most mistakes and retake the test afterwards.